

# No Evidence of Benefit from Marchenko–Pastur Spectral Gradient Filtering on Noisy Financial Time Series: A Pre-Registered Negative Result with a Target-Independence Proof

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## Abstract

Per-sample gradient “consensus” filtering—eigendecomposing the  $B \times B$  inter-sample gradient similarity matrix and keeping only directions above a Marchenko–Pastur-style threshold—has shown strong label-noise robustness in classification. We ask whether this mechanism reduces the effect of noise when training on large, low signal-to-noise financial data. Under a fully pre-registered protocol on the Numerai v5 dataset (era-purged tuning/verdict split, frozen thresholds, moving-block-bootstrap inference, compute-matched tuning), the answer is negative: we find *no evidence of benefit under the affordable tuning budget*, and at the pre-registered operating point the filter significantly degraded out-of-sample performance on both architectures tested (MLP:  $-0.00527$  mean per-era correlation, 95% CI  $[-0.00886, -0.00181]$ ; GRU:  $-0.00484$ , CI  $[-0.00758, -0.00211]$ , against a  $+0.0064$  baseline edge). Two mechanistic contributions explain why the label-noise story does not transfer. First, we prove that for scalar-output MSE the filter’s engagement eigenspectrum is exactly target-independent: per-sample gradients reduce to  $\pm(\text{residual sign}) \times \text{a target-independent Jacobian direction}$ , so “consensus” measures feature/factor structure, not signal (verified in fp64 to  $8.9 \times 10^{-16}$ ; independently re-verified). Second, matched controls show MP-eigenselection is statistically indistinguishable from a norm- and  $k$ -matched *random*-subspace projection on both architectures. An independent audit re-executed the main experiment with fresh seeds and an independently constructed evaluation shard and reproduced the result. Claims are scoped to a subsampled, low-edge regime.

## 1 Introduction

- **Motivating question.** Can the Spectral Optimizer—a gradient-filtering wrapper around AdamW that eigendecomposes the  $B \times B$  per-sample gradient similarity matrix and keeps only directions many samples agree on (Marchenko–Pastur-style threshold)—reduce the effect of noise when training on large, noisy financial time-series data? Concretely: does it improve out-of-sample predictive performance relative to tuned AdamW, and is any effect consistent across architectures?
- **Why the outcome was genuinely uncertain.** In prior (unpublished) work the same filter delivered strong label-noise robustness on MNIST at 90% label corruption ( $\sim 80\%$  accuracy vs.  $\sim 37\%$  for Adam), characterized as a “coherence amplifier”. But the same work found it hurts when signal is weak and not yet gradient-dominant, and Feldman’s long-tail memorization theory [6] predicts consensus filtering could suppress rare genuine signal in heavy-tailed financial data. Prior evidence pointed in both directions.
- **Design.** A pre-registered, fail-fast evaluation on the Numerai v5 tournament dataset (large, obfuscated, low-SNR financial features with era structure): era-purged tuning/ver-

dict block separation with embargo, a frozen three-way verdict rule (helps / doesn't help / hurts) with a pre-registered practical-significance threshold, moving-block-bootstrap inference, compute-matched tuning budgets [22], mechanism controls, and engagement diagnostics on every run. Five experiments, all on single L40 GPUs, ~15 GPU-minutes total.

- **Headline result (negative).** Under the pre-registered protocol the reportable claim is *no evidence of benefit under the affordable tuning budget* (the spectral arm afforded ~1 compute-matched tuning trial vs. AdamW's 12). The raw numbers behind that downgraded wording: the filter significantly *degraded* out-of-sample performance on both architectures (MLP:  $-0.00527$  mean per-era `numera1_corr`, 95% CI  $[-0.00886, -0.00181]$ ; GRU:  $-0.00484$ , CI  $[-0.00758, -0.00211]$ ), erasing ~82–83% of a  $+0.0064$  baseline edge.
- **Co-primary contribution 1: a target-independence proof.** For scalar-output MSE, the filter's engagement eigenspectrum is *exactly* target-independent at fixed parameters: row normalization reduces each per-sample gradient to  $\pm(\text{residual sign}) \times$  a target-independent Jacobian direction, so permuting targets conjugates the similarity matrix by an orthogonal sign matrix (verified in fp64 to  $8.9 \times 10^{-16}$ ; independently re-verified at  $2.1 \times 10^{-14}$  with scoping controls showing the theorem is non-vacuous). "Engagement" therefore cannot certify that the mechanism found target signal on this task class.
- **Co-primary contribution 2: matched mechanism controls.** MP-eigenselection is statistically indistinguishable from a norm- and  $k$ -matched *random*-subspace projection on both architectures (MLP:  $-0.00116$ , CI  $[-0.00366, +0.00134]$ ; GRU:  $-0.00011$ , CI  $[-0.00233, +0.00215]$ ; equivalence at  $\pm 0.004$  resolution). All three filtered arms (spectral, random-subspace, sign-agreement) hurt similarly: the harm tracks agreement-style update modification as a class, not the spectral machinery.
- **Robustness.** An independent results audit re-executed the load-bearing experiment on fresh seeds {10, 11, 12} and an independently constructed evaluation shard; the headline reproduced in category, direction, and magnitude, as did the spectral-vs-random null (6-seed pooled headline  $-0.00537$ , CI  $[-0.00853, -0.00222]$ ).
- **Framing.** The contribution is the transfer verdict and its mechanistic explanation, not a new optimizer. A well-supported negative result was priced as a full deliverable from the outset; this paper reports one, with its scope limits stated plainly (subsampling, low-edge regime; single spectral operating point).

## 2 Related Work

The literature search underlying this section was conducted on 2026-07-31 across academic, lab-blog, and community channels (28 curated references). No discovered work combines (a) per-sample gradient covariance eigendecomposition with a random-matrix-theory threshold as the training update rule with (b) evaluation on low-SNR financial prediction; the novelty claim rests on absence of evidence in an actively producing niche and is positioned against the nearest neighbor below rather than claimed as a vacuum.

**Per-sample gradient filtering (2024–2026).** The closest empirical precedent is Gradient Agreement Filtering [4], which filters or reweights macrobatch updates by cross-microbatch gradient agreement using pairwise orthogonality/correlation checks, improving robustness under image-classification label noise. The Spectral Optimizer differs in both mechanism (full  $B \times B$  similarity-matrix eigendecomposition with a Marchenko–Pastur-style threshold, rather than pairwise checks) and domain (low-SNR financial regression, which GAF never touched). A

burst of adjacent work confirms per-sample gradient statistics are a live optimizer-design surface: orthogonal-gradient constraints against noisy-label memorization [13], per-sample clipping with heavy-tailed convergence theory [16], gradient spectral entropy for backdoor-sample detection [28], and per-sample momentum with low-pass filtering for DP-SGD [26]. None uses covariance-eigenspectrum thresholding as the update rule.

**Why consensus filtering might work—and why it might not.** The theoretical ancestor is the Coherent Gradients hypothesis [3]: SGD generalizes because gradient directions shared across many examples are reinforced more than idiosyncratic ones—an intuition independently articulated in the community [20], and related to broader treatments of the memorization–generalization spectrum [24]. The sharpest counter-hypothesis is Feldman [6]: memorization of rare long-tail examples can be *necessary* for generalization, so a consensus filter could suppress rare genuine signal in heavy-tailed financial data. The noisy-label literature operates mostly at the loss/data level [11, 12, 23]; optimizer-level gradient-covariance filtering is a comparatively unexplored corner of that space.

**Random matrix theory in finance.** Marchenko–Pastur eigenvalue cleaning of financial *return* correlation matrices is a mature,  $\sim 20$ -year quantitative-finance literature [2, 10, 19, 21]: large fractions of empirical eigenspectra are indistinguishable from noise, and clipping them improves portfolio risk estimates. The Spectral Optimizer transposes this denoising logic from returns covariance (a post-hoc estimation step) to per-sample *gradient* covariance during training; the transposition appears unpublished. Our results delimit it: on this task the analogy breaks because the gradient similarity spectrum turns out to be provably target-independent (Section 4.2).

**The baseline is not naive.** Gradient noise in deep learning is structured and heavy-tailed [5, 25], and sign-based/adaptive normalization—what Adam already does—confers meaningful robustness in exactly this regime [27]; robust adaptive variants date back to at least Gulcehre et al. [9]. A 2026 benchmark of 15 optimizers on tabular MLPs finds Muon consistently beats AdamW [8], so “beats default-hyperparameter Adam” would be a strawman claim. We therefore adopt the codified fair-comparison methodology: matched tuning budgets, multiple seeds, standardized reporting [1, 22], with tuned AdamW as the primary opponent and the filter-on/filter-off ablation at identical base configuration as the cleanest mechanism isolation. As a contrast at the mechanism level, the gradient noise scale [15] treats gradient noise as a scalar SNR statistic for batch-size selection, and K-FAC-style methods model gradient covariance for curvature preconditioning [14]—neither performs noise-rejection filtering of the update.

**Evaluation protocol for Numerai-style data.** No academic Numerai literature was found; the practitioner-standard protocol comes from community sources: era-based purged cross-validation with embargo, per-era correlation as the metric [18], and feature-neutral correlation to separate genuine signal from feature overfitting [17]. MLPs remain standard and competitive for tabular data [7]. One structural constraint shaped our design: Numerai asset identifiers reset each era, so a longitudinal recurrent arm is impossible on this dataset; the sequence arm uses a within-era framing (Section 4.6).

## 3 Methodology

### 3.1 The Spectral Optimizer

The object under test is fixed in writing (amendment F11): the `SpectralConsensusFilter`, the exact  $B \times B$  variant from the prior project’s `spectral_optimizer.py`, reused byte-identically

(verified), not re-implemented. At each step it (i) computes per-sample gradients  $g_1, \dots, g_B$  via `torch.func vmap(grad)`; (ii) row-normalizes them and forms the  $B \times B$  similarity matrix  $S_{ij} = \hat{g}_i^\top \hat{g}_j$ ; (iii) eigendecomposes  $S$  and keeps eigendirections above a Marchenko–Pastur-style threshold (registered *hard* mode: eigenvalues above  $2 \cdot \text{trace}(S)/B$ ); (iv) projects the uniform sample-weight vector onto the kept eigensubspace and applies the resulting weights to the *unnormalized* gradients to form the update passed to AdamW. Because anti-aligned (cancelling) components can be discarded, the filtered update norm is not bounded by the mean-gradient norm—the filter can amplify as well as shrink, which turns out to matter (Section 4.5).

### 3.2 Data and era-purged protocol

We use the public Numerai v5.0 dataset (obfuscated financial features, weekly eras,  $\sim 20$ -day forward-return target): 705 “medium” features, per-sample MSE on the centered target. To fit the compute profile, training uses a subsampled shard of 384,000 rows (150 eras, 0425–0574,  $\leq 2,560$  rows/era), identical across all arms. The validation-side eras are split once, in advance (amendment F1), into a **tuning block** (eras 0579–0966; 388 eras), an **embargo** (0967–0970; 4 eras =  $\lceil 20 \text{ d} / 5 \text{ d} \rceil$ , amendment F5), and a **verdict block** (0971–1225; 255 eras). All tuning, mode selection, and peeking touch only the tuning block; the verdict block was evaluated exactly once, after all arms and configurations were frozen (corroborated by git commit ordering). Sanity and leakage bands were recalibrated from current example-model statistics (amendment F9): sanity band  $[0.0087, 0.0522]$  mean per-era `numerai_corr`, leakage gate  $> 0.0696$ . The metric is per-era `numerai_corr` (with Spearman as a variant), following practitioner protocol [17, 18].

### 3.3 Pre-registered verdict machinery

The deliverable is a three-way verdict with teeth:

- **Helps / hurts:** the paired per-era CI on the filter-on – filter-off difference excludes zero *and* the point estimate exceeds a practical-significance threshold  $F3$ , defined as the minimum of 0.005 and  $0.25 \times$  the realized tuned-baseline mean per-era corr, and frozen before unblinding at **0.00398** (from the 3-seed, winner’s-curse-corrected baseline mean  $+0.01594$ ).
- **Doesn’t help:** an equivalence-style claim—the CI is narrow enough to exclude improvements  $\geq F3$ —not  $p > 0.05$ .
- **Inference:** per-era scores are autocorrelated (measured lag-1 acf =  $+0.25$ ; the target horizon spans  $\sim 4$  eras), so all CIs use a paired moving-block bootstrap with block length  $L = 4$  and 10,000 resamples (amendment F5). A corr-Sharpe secondary route uses the identical machinery (amendment F10).
- **Tuning-adequacy downgrade (amendment F12):** tuning budgets are matched in *compute*, not trial count. AdamW received a 12-trial sweep ( $\sim 44$ s); the same compute affords  $\sim 1$  spectral trial (63s/run), so the spectral arm ran at priors transferred from the prior project. This asymmetry was recorded before unblinding as an under-exploration signature, with a binding rule: a “hurts” verdict must be downgraded to “*no evidence of benefit under the affordable tuning budget*”, reported with the raw numbers alongside. That rule fired, and the downgraded wording is used throughout this paper.
- **Engagement diagnostics:** every run logs the number of kept eigendirections  $k$ , kept-energy fraction, update-norm ratio, and cosine diagnostics, so a null reads “mechanism engaged but did not help”, not “implementation no-op”.

| Component                                  | $P_{\text{succ}}$ | $T$ (h) | $\lambda$ | Outcome                  |
|--------------------------------------------|-------------------|---------|-----------|--------------------------|
| Spectral engagement on financial gradients | 0.40              | 0.75    | 1.22      | FAIL on C2 (proof; §4.2) |
| Sequence-arm vmap feasibility (GRU)        | 0.35              | 1.0     | 1.05      | PASS                     |
| Statistical power of paired per-era design | 0.60              | 0.5     | 1.02      | PASS                     |
| Design fits 5-experiment / 30-min budget   | 0.80              | 0.25    | 0.89      | PASS                     |
| OHLCV fallback dataset for sequence arm    | 0.80              | 0.5     | 0.45      | not needed               |
| Regression integration of the filter       | 0.75              | 0.75    | 0.38      | PASS                     |
| vmap throughput on L40 within job cap      | 0.85              | 0.5     | 0.33      | PASS                     |
| Baseline sanity: tuned AdamW in corr band  | 0.70              | 1.5     | 0.24      | PASS                     |
| Numerai data + era-purged protocol         | 0.85              | 1.0     | 0.16      | PASS                     |

Table 1: Steinhart-style decomposition with fail-fast ordering by  $\lambda$ . The highest- $\lambda$  component—whether spectral “engagement” is meaningful on financial gradients—failed its C2 conjunct by proof, converting to the paper’s co-primary mechanistic finding rather than aborting the run (the pre-registered hard degeneracy trigger did not fire).

### 3.4 Fail-fast decomposition

The plan was decomposed into nine components ranked by  $\lambda = -\ln(P_{\text{success}})/T_{\text{hours}}$  (information gained per unit time from attacking the riskiest cheap components first), executed as five experiments packed along the dependency structure. Table 1 shows the components with their pre-registered  $P_{\text{success}}$  estimates and outcomes.

Before execution, three independent adversarial reviews (assumption analysis, mentor review, pre-mortem) produced 13 binding protocol amendments (F1–F13: era separation, frozen thresholds, block-bootstrap inference, compute-matched tuning with the downgrade rule, conditional cut order protecting mechanism controls, and claim-naming rules) and 5 diagnostic additions (C1–C5), all folded into the plan before any comparison was unblinded. The most consequential additions were C2 (a permuted-target null-spectrum control) and C4 (a random-subspace control matched to the measured kept- $k$  and update-norm trajectories), which between them produced both mechanistic contributions.

### 3.5 Models, arms, and compute

**Architectures.** (i) MLP 705–256–64–1 (197k parameters); (ii) a hand-rolled functional single-layer GRU (hidden 64, 21.8k parameters, vmap-compatible; sequence-to-one head) consuming each row’s 705 features reshaped to a  $T=15 \times D=47$  within-era sequence. Both use within-era minibatches of  $B=1024$  (amendment F6), 2000 steps, base configuration  $\texttt{t07}$  (lr  $10^{-3}$ , wd  $10^{-3}$ , dropout 0.2) selected by the AdamW sweep on tuning-block eras only.

**Arms** (identical shard, seeds, and data order): *filter-off* (tuned AdamW); *filter-on* (SpectralConsensusFilter, hard mode, mp\_factor 2.0—registered from engagement diagnostics only, amendment F2); *C4 random-subspace control* (random orthonormal sample-subspace with  $k(t)$  and update-norm-ratio( $t$ ) matched step-by-step to the same-seed filter-on run); a *GAF-style sign-agreement ablation* (per-coordinate  $|\sum_i \text{sign}(g_i)| \geq 2\sqrt{B}$ ; a sign-agreement variant, not the exact microbatch-cosine method of Chaubard et al. [4]); and a *zero-predictor* sanity baseline. Three seeds per arm (extended to six on the headline by the audit re-execution, Section 4.7).

**Compute.** All GPU work ran on single NVIDIA L40s (48 GB) on a shared Slurm cluster’s free compute partition, every job under 10 minutes; total producer GPU spend  $\sim 15$  minutes across four jobs, plus  $\sim 9$  minutes for the audit re-execution. Analysis ran on local CPU.

| Comparison                                                                                           | Estimate | 95% CI (MBB, $L=4$ ) | Reading                          |
|------------------------------------------------------------------------------------------------------|----------|----------------------|----------------------------------|
| <i>Headline: filter-on – filter-off (vs. tuned AdamW)</i>                                            |          |                      |                                  |
| MLP, seeds 0–2 (exp-004)                                                                             | −0.00527 | [−0.00886, −0.00181] | hurts → F12 downgrade            |
| GRU, seeds 0–2 (exp-005)                                                                             | −0.00484 | [−0.00758, −0.00211] | hurts → F12 downgrade            |
| MLP re-run, seeds 10–12, producer shard                                                              | −0.00546 | [−0.00892, −0.00204] | reproduces                       |
| MLP re-run, seeds 10–12, audit shard                                                                 | −0.00562 | [−0.00911, −0.00201] | reproduces                       |
| MLP pooled, 6 seeds                                                                                  | −0.00537 | [−0.00853, −0.00222] | reproduces                       |
| <i>Mechanism control: filter-on – <math>C_4</math> random subspace (norm/<math>k</math>-matched)</i> |          |                      |                                  |
| MLP (exp-004)                                                                                        | −0.00116 | [−0.00366, +0.00134] | null (equiv. at $\pm 0.004$ )    |
| GRU (exp-005)                                                                                        | −0.00011 | [−0.00233, +0.00215] | null                             |
| MLP re-run, producer shard                                                                           | +0.00054 | [−0.00161, +0.00275] | null                             |
| MLP re-run, audit shard                                                                              | −0.00114 | [−0.00334, +0.00108] | null                             |
| <i>Secondary: corr-Sharpe diff (filter-on – filter-off)</i>                                          |          |                      |                                  |
| MLP                                                                                                  | −0.200   | [−0.416, −0.004]     | marginal (bootstrap-RNG fragile) |
| GRU                                                                                                  | −0.242   | [−0.414, −0.073]     | robust                           |

Table 2: Headline results: mean per-era `numera1_corr` differences over the 255 verdict-block eras (3-seed means unless noted; frozen practical-significance threshold  $F3 = 0.00398$ ). “F12 downgrade”: reportable claim is *no evidence of benefit under the affordable tuning budget* (Section 3.3). Realized baseline (filter-off) edge: +0.00641 (MLP), +0.00586 (GRU)—below the tuning-block sanity band (regime drift; Section 6).

## 4 Experiments and Results

### 4.1 Headline results

Table 2 and Figure 1 summarize the run’s findings. Under the pre-registered protocol the verdict machinery returned the category *hurts* on both architectures; the binding F12 downgrade rule (Section 3.3) converts the reportable claim to **no evidence of benefit under the affordable tuning budget**, with the raw numbers reported alongside. The load-bearing mechanism control is null on both architectures: MP-eigenselection is statistically indistinguishable from a norm- and  $k$ -matched random-subspace projection (equivalence at  $\pm 0.004$  resolution). The independent audit re-execution (fresh seeds, independently constructed evaluation shard) reproduces both findings.

### 4.2 Exp-001: gate bundle and the target-independence result

The first experiment bundled the infrastructure gates (integration smoke, data/protocol construction, throughput) with the run’s highest- $\lambda$  question: is spectral “engagement” on financial gradients meaningful? The infrastructure gates all passed (filter-off reproduces plain AdamW to fp64 precision; 31 ms/step filter-on at  $B=1024$  on one L40; 643 usable validation eras). The headline engagement gate is logged as a **FAIL on its C2 conjunct**—a pre-registered pass criterion requiring real-target and permuted-target engagement spectra to be distinguishable—and this failure is the run’s first mechanistic contribution.

Mechanically, the filter is healthy and selective on real financial gradients: for every threshold mode  $\times$  batch size  $\times$  batch composition,  $0 < k < B$  on 100% of logged steps ( $k$  settling at 3–7), kept-energy fraction 0.29–0.89, and the C3 cosine diagnostics show the filtered update is not mean-gradient smoothing (median cosine 0.25–0.83; the  $> 0.95$  degeneracy pivot never fired). The pre-registered *hard* fail-fast trigger therefore did not fire. But the C2 permuted-target control was indistinguishable from real targets—provably so:

**Theorem 4.1** (Target independence of the engagement spectrum). *Let  $f_\theta$  be a scalar-output model trained with per-sample MSE, so the per-sample gradient is  $g_i = 2(f_\theta(x_i) - y_i)\nabla_\theta f_\theta(x_i)$ . After row normalization,  $\hat{g}_i = \text{sign}(f_\theta(x_i) - y_i)\nabla f_i/\|\nabla f_i\|$ . Hence for any targets  $y'$  at the*

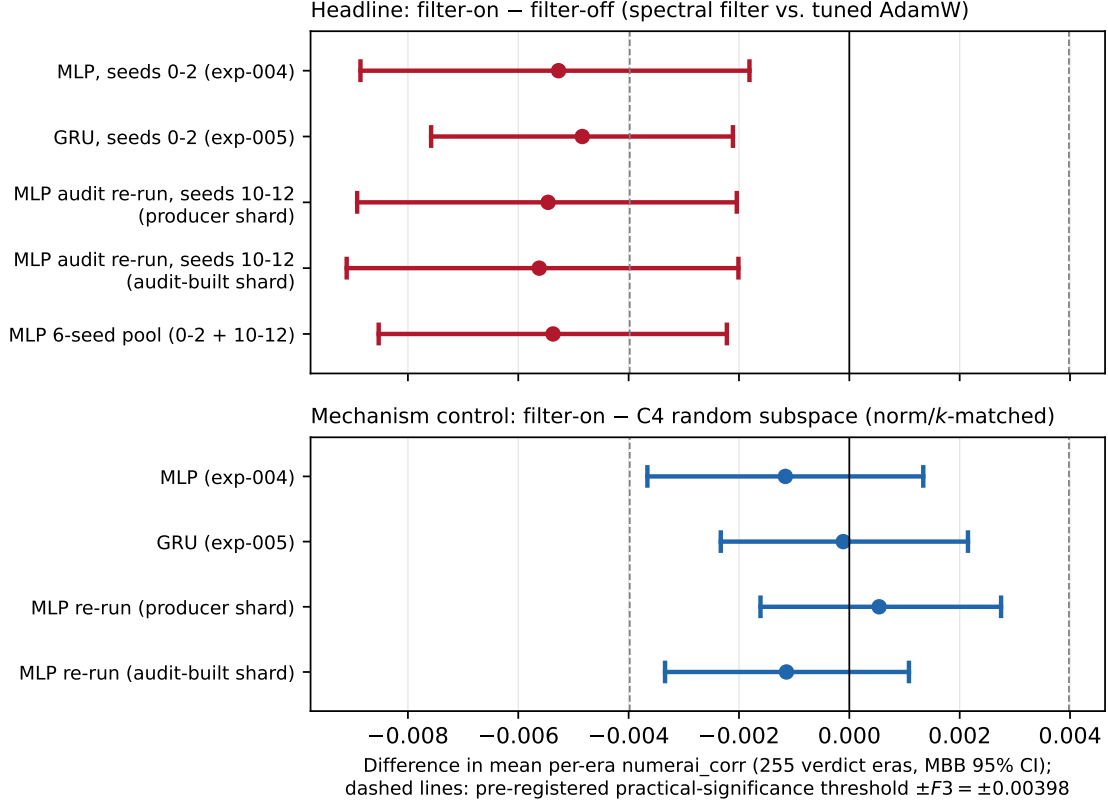


Figure 1: Forest plot of the run’s principal paired comparisons (mean per-era `numera_i_corr` differences over 255 verdict eras, moving-block-bootstrap 95% CIs). Top: the filter significantly degrades out-of-sample performance vs. tuned AdamW on both architectures, and the result reproduces under the audit’s fresh-seed re-execution on two independently constructed evaluation shards. Bottom: the spectral filter is indistinguishable from its norm/ $k$ -matched random-subspace control everywhere.

same parameter point, the similarity matrix satisfies  $S(y') = D S(y) D$  with  $D = \text{diag}(\pm 1)$  orthogonal. The eigenspectrum of  $S$ —and therefore  $k$ , the MP-threshold count, the kept-energy fraction, and every spectrum-derived engagement diagnostic—is exactly target-independent at fixed parameters.

The proof was verified numerically in fp64 (max eigenvalue difference  $8.9 \times 10^{-16}$  under target permutation;  $1.3 \times 10^{-15}$  under entirely different targets) and independently re-verified by the results audit with a different construction ( $2.1 \times 10^{-14}$ ), together with two scoping controls showing the theorem is non-vacuous and correctly scoped: the *unnormalized* spectrum is strongly target-dependent (difference  $\approx 61$ ), and a 2-output MSE head breaks the invariance (difference  $\approx 0.61$ ). Targets influence the spectrum only through the parameter trajectory, which at this horizon/SNR does not measurably diverge: late-training real-vs-permuted diagnostic distributions are statistically indistinguishable (the two nominally significant tests out of twelve point in opposite directions), and end-of-run training loss on real targets is not below permuted-target loss.

Two scoping notes. First, the update *direction* does depend on targets (through residual signs and magnitudes); what fails is the ability of the engagement diagnostics to certify that the mechanism found target signal—the premise for interpreting any engagement-gated verdict. Second, the theorem is specific to scalar-output per-sample losses with row normalization; multi-class cross-entropy gradients (the prior project’s MNIST label-noise setting) are *not* scalar

multiples of a target-independent vector, which cleanly explains why the same diagnostics were informative there and are not here. A synthetic calibration test (C1) exhibits the failure mode directly: in hard mode the filter correctly discards i.i.d. noise and keeps a planted spike, but also keeps a correlated *zero-signal* confound at cosine 0.96 to the confound direction—and Theorem 4.1 says the real data is the confound case, fully. Two additional observations from this gate recur later: the similarity spectrum undergoes strong target-independent rank collapse during training (one direction accumulating  $\sim 80\text{--}90\%$  of the mass), and the filter frequently *amplifies* the update (norm ratio  $> 1$  on 25–70% of steps)—it is not shrinkage.

Following the pre-registered fail semantics, the run proceeded to the comparison under an amended interpretation frame recorded before any unblinding: the proof is a co-primary deliverable; any verdict is attributed to target-blind spectral-subspace regularization, never “consensus on signal”; and the C4 random-subspace control is load-bearing. Binding registrations made at this gate from diagnostics only: threshold mode *hard* (mp\_factor 2.0), within-era batch composition.

### 4.3 Exp-002: sequence-arm feasibility

A hand-rolled functional GRU cell was validated for per-sample gradients under `vmap(grad)`: element-wise agreement with a per-sample loop reference to  $1.7 \times 10^{-6}$  (tolerance  $10^{-5}$ ) at  $B=64$ ; a full filtered step took 1.17s on CPU at  $B=1024$ , projecting well under the job cap on an L40. This authorized the architecture-consistency arm (exp-005) rather than an MLP-only fallback. The correctness assert was re-run on-cluster at job start (max difference  $4.8 \times 10^{-7}$ ).

### 4.4 Exp-003: baseline sanity, power, and the go/no-go gate

A 12-trial AdamW sweep on tuning-block eras only produced best configuration `t07` with mean per-era `numera_i_corr` +0.0196 (3-seed mean +0.01594; winner’s-curse shrinkage  $\sim 19\%$ ), inside the recalibrated sanity band; the zero-predictor was  $\approx 0$  and the leakage gate did not fire. From these outputs the verdict threshold was frozen at  $F3 = 0.00398$ , the block length at  $L=4$  (measured lag-1 autocorrelation +0.25), and a moving-block-bootstrap power simulation of the paired 3-seed design at 255 verdict eras gave a median CI half-width of 0.00219 with  $P(\text{some verdict category reachable}) = 0.910 \geq 0.6$  (go/no-go gate: GO; detection probability  $\approx 0.90\text{--}0.92$  at  $\pm F3$ ). Two under-exploration signatures were recorded here, before unblinding, and bind the wording of any negative verdict: the baseline’s best weight-decay and dropout sit at the grid boundary with improving trends (the baseline itself may be slightly under-tuned—fixing this would strengthen the negative), and the compute-matched budget affords the spectral arm only  $\sim 1$  tuning trial vs. AdamW’s 12 (the asymmetry that mandates the F12 downgrade).

### 4.5 Exp-004: the verdict experiment (MLP)

The first and only unblinding of the 255 verdict eras, with all arms and configurations frozen beforehand. Table 3 gives arm levels; the paired headline is in Table 2. All three per-seed headline differences are negative (seed-2 CI touches zero:  $[-0.01069, +0.00023]$ ); direction is seed-consistent.

**Mechanism attribution (the load-bearing control).** The spectral filter is statistically indistinguishable from its norm- and  $k$ -matched random-subspace control ( $-0.00116$ , CI  $[-0.00366, +0.00134]$ )—an equivalence claim at the  $\pm 0.004$  resolution the design affords, not an identity claim. All three filtered arms hurt by similar amounts ( $-0.0041$  to  $-0.0068$  vs. baseline) despite very different mechanisms (eigenspectral, random-subspace, coordinate sign-agreement): per-sample-agreement-style update modification *as a class* degrades out-of-sample performance in this regime. Combined with Theorem 4.1, the account is closed from both ends:

| Arm                      | numera_i_corr | Spearman | corr-Sharpe | per-seed nc                 |
|--------------------------|---------------|----------|-------------|-----------------------------|
| filter-off (tuned AdamW) | +0.00641      | +0.00698 | 0.309       | +0.0053 / +0.0066 / +0.0073 |
| filter-on (spectral)     | +0.00113      | +0.00090 | 0.109       | +0.0006 / +0.0007 / +0.0021 |
| C4 random-subspace       | +0.00229      | +0.00112 | 0.178       | +0.0004 / +0.0022 / +0.0043 |
| GAF-style sign-agreement | −0.00042      | +0.00004 | −0.032      | −0.0015 / +0.0012 / −0.0009 |
| zero-predictor           | +0.00063      | +0.00057 | —           | sanity $\approx 0$          |

Table 3: Exp-004 arm levels (mean per-era scores over 255 verdict eras, 3-seed means, MLP).

the selection is target-blind by proof, and its effect is no different from projecting onto a random sample-subspace with the same kept- $k$  and norm trajectory.

**Engagement during the verdict runs.** The filter did real, selective work throughout (selective on 100% of logged steps):  $k$  collapsed from  $\sim 52$  at step 0 to a median of  $\sim 1$ , with update-norm ratio  $\sim 9.5\text{--}13.9\times$  the mean gradient—the filter effectively replaced the averaged gradient with a single amplified dominant eigendirection. The C4 control matched exactly this ( $k$ , ratio) trajectory, which is what makes the null comparison meaningful. The a-priori plausible “implicit regularization by norm shrinkage” story is inverted here: the filter amplifies.

**Tail-era breakdowns.** Bucketing eras by the baseline’s own realized per-era corr shows a strong monotone gradient (filter “helps” on the baseline’s worst quartile, +0.019; hurts most on its best, −0.030), but this split is outcome-conditioned and mechanically contaminated by regression to the mean (flagged in the analysis code before results); we report it as descriptive only. The evidentially meaningful, unconditioned split by per-era *target dispersion* shows roughly uniform harm across quartiles—no evidence of the Feldman-style pattern [6] in which harm concentrates in high-dispersion (tail) eras. The corr-Sharpe also fell, so the filter did not buy stability either.

**Sanity notes.** The realized verdict-block baseline (+0.00641) sits *below* the tuning-block sanity band [0.0087, 0.0522]; the leakage gate did not fire. The verdict eras are a genuinely harder, later regime (the example model also drops from 0.0418 to 0.0235 between blocks)—this regime drift scopes the claim (Section 6) and made the frozen threshold a  $\sim 62\%$  relative bar rather than the intended 25%; the verdict does not depend on it (the CI excludes zero regardless).

#### 4.6 Exp-005: architecture consistency (GRU)

The same data, shards, seeds, frozen thresholds, and inference machinery, with the only change being the architecture (within-era Numerai framing,  $15 \times 47$  reshape; the supported claim is *architecture consistency*, not “second setting”). The GRU reproduces the MLP’s result: headline −0.00484, CI [−0.00758, −0.00211] (same category, F12 downgrade applied identically); spectral-vs-random null replicates even more tightly (−0.00011, CI [−0.00233, +0.00215]); corr-Sharpe −0.242, CI [−0.414, −0.073] (robust); the dispersion split again shows no tail pattern. All six per-seed headline differences across both experiments are negative.

Notably, the filter engages in a *different regime* on the GRU ( $k \approx 60$  kept directions, mild attenuation at norm ratio 0.80–0.85) than on the MLP ( $k \approx 1$ ,  $\sim 10\times$  amplification), yet produces near-identical harm and the same null against the random-subspace control—strengthening the account that the harm tracks the generic subspace-projection/rescaling intervention class rather than any particular spectral selection or amplification behavior. The GRU arms ran at the  $\tau 07$  configuration transferred from the MLP (frozen pre-unblinding and identical across arms, so the paired comparison is protocol-valid; the skipped GRU sweep is a recorded limitation, Section 6). The untuned GRU baseline (+0.00586) landed within  $\sim 9\%$  of the tuned MLP baseline despite  $9\times$  fewer parameters.

## 4.7 Robustness: independent audit re-execution

An independent results audit re-executed the load-bearing exp-004 on the cluster using the producer’s frozen code with **fresh seeds {10, 11, 12}** and, additionally, an **auditor-built verdict shard** subsampled independently from the raw validation parquet (different RNG; the producer’s shard was also row-matched against the raw parquet, ruling out fabrication). The headline reproduces in category, direction, and magnitude on both shards ( $-0.00546$  and  $-0.00562$  vs. the original  $-0.00527$ ), the 6-seed pooled headline is  $-0.00537$ , CI  $[-0.00853, -0.00222]$ , and the C4 spectral-vs-random null replicates (Table 2; all audit CIs from independent bootstrap code). The audit also independently re-verified Theorem 4.1 and re-derived every reported number from primary artifacts (robustness cross-checks: paired  $t$ -test  $p = 1.5 \times 10^{-4}$ , Wilcoxon  $p = 1.8 \times 10^{-4}$ ).

The re-execution surfaced one implementation defect that the original runs did not hit: under the rank-collapse regime this study itself documents (one eigenvalue carrying  $\sim 80\text{--}90\%$  of the mass), `fp32 linalg.eigh` on GPU can fail to converge, crashing the run stochastically by seed. The original 6-of-6 producer runs completed without encountering it; the audit’s mathematically equivalent CPU-fp64 fallback (which fired on 4 of 18,000 filtered steps) completed the re-run and reproduced all conclusions. This is an availability defect of the released implementation, not a bias in the reported numbers; it is disclosed in Section 6.

## 5 Discussion

**The answer to the motivating question is no—with a precise scope.** On this large, noisy financial dataset, under a pre-registered protocol with matched compute budgets, the Spectral Optimizer did not reduce the effect of noise: there is no evidence of benefit under the affordable tuning budget, and at the pre-registered operating point the filter significantly degraded out-of-sample performance on both an MLP and a GRU, erasing  $\sim 82\text{--}83\%$  of a small ( $+0.0064$ ) baseline edge. The result survived an adversarial audit that re-executed the experiment on fresh seeds and an independently constructed evaluation split. The claim is scoped to a subsampled, low-capacity pipeline whose verdict-block edge sits below the practitioner sanity band, and to a single spectral operating point.

**Why the label-noise story does not transfer.** The two mechanistic results give a consistent account. By Theorem 4.1, on scalar-output MSE the quantity the filter thresholds—the eigenspectrum of the normalized per-sample gradient similarity matrix—is exactly target-independent at fixed parameters. “Directions many samples agree on” are, on this task class, directions of the *feature/Jacobian factor structure*, not of target signal; the coherence-amplifier interpretation from the label-noise setting has nothing to grip. Cross-entropy classification escapes the theorem (per-sample gradients are not scalar multiples of a target-independent vector), which explains why the same diagnostics were informative on MNIST label noise and are not here. Empirically, the C4 control closes the loop from the other end: conditioning on the kept- $k$  and update-norm trajectories, replacing MP-eigenselection with a random sample-subspace changes nothing detectable ( $\pm 0.004$  resolution), on either architecture, in either engagement regime ( $k \approx 1$  with  $10\times$  amplification on the MLP;  $k \approx 60$  with mild attenuation on the GRU).

**The harm is a class effect, not a spectral effect.** All three agreement-style arms—spectral, random-subspace, and sign-agreement—hurt by similar amounts. Whatever damage is done appears attributable to per-sample-agreement-style update restriction/rescaling itself. Two candidate stories for *why* such interventions hurt here are already constrained by the data: it is not a stability trade (corr-Sharpe fell, robustly so on the GRU), and it is not a Feldman-style tail effect (the unconditioned dispersion split shows uniform harm, not harm

concentrated in high-dispersion eras). A plausible remaining account—that in a regime where the mean gradient is the only weakly informative direction, replacing it with an amplified low-rank projection of factor structure simply injects variance where there is no exploitable coherent signal—is consistent with the prior project’s own weak-signal failure mode, but our design does not isolate it; we flag it as interpretation, not finding.

**On the strength of the negative.** The pre-registered machinery was designed so that a negative would be meaningful: the F4 power gate showed all three verdict categories reachable (power 0.91 at the frozen threshold) before any slot was spent; engagement diagnostics confirm the mechanism did sustained, selective work during every verdict run; and the audit classified the outcome as an adequately powered honest negative (exit reason: *true-null*) rather than an artifact. The main interpretive guard is deliberate and one-sided: because the spectral arm afforded only  $\sim 1$  compute-matched tuning trial, “hurts” cannot be distinguished from “under-tuned spectral arm” at this budget, so the paper’s claim stays at “no evidence of benefit under the affordable tuning budget” with the raw numbers alongside. The mirrored signature (the baseline’s regularization grid-boundary trend) cuts the other way: a better-tuned baseline would, if anything, widen the gap.

**What would change our mind.** A demonstration that a tuned spectral configuration (wider operating-point sweep at a declared, enlarged, matched budget) recovers or beats the baseline in this regime; or a full-scale (non-subsampled) evaluation in which the verdict-block edge returns to the practitioner band and the effect reverses. Both are concrete, costed follow-ups in Section 7.

## 6 Limitations

An independent results audit reviewed every claim in this paper against primary artifacts, re-executed the load-bearing experiment, and exited with disposition *honest-negative* (audit exit reason: **true-null**—an adequately powered genuine negative, reproduced under fresh-seed re-execution on an independently constructed evaluation split, with no claim-flipping defect found). The audit’s eight unresolved-for-write-up findings follow, each with its triage disposition; then design-level limitations from the pre-mortem triage.

1. **Regime drift / below-band baseline.** The verdict-block baseline (+0.0064) sits below the pre-registered practitioner sanity band [0.0087, 0.0522]; the frozen threshold consequently realized as a  $\sim 62\%$  relative bar rather than the intended 25%. All claims are scoped to this low-edge, subsampled regime; the baseline still carries real signal (per-era  $t \approx 4.9$  vs. the zero-predictor) and the CI-excludes-zero verdict is threshold-independent. *Disposition: deferred to future work*—a full-scale non-subsampled verdict needs 10–20 L40-hours or A100-class compute, beyond this run’s free-partition, sub-30-minute-job profile (item 1, Section 7).
2. **Tuning-affordability asymmetry (F12).** The spectral arm afforded  $\sim 1$  compute-matched tuning configuration vs. AdamW’s 12 trials, so “hurts” cannot be separated from “under-tuned spectral arm”. *Disposition: addressed by wording*—the pre-registered downgrade rule binds the claim to “no evidence of benefit under the affordable tuning budget” everywhere in this paper, with raw numbers alongside; a declared enlarged-budget sweep is future work (item 2).
3. **GRU configuration not tuned.** The GRU arms ran the MLP’s `t07` configuration under a fallback whose premise (a wiped tuning shard) was later found false on the worker node; the configuration was frozen pre-unblinding and identical across arms, so the paired

comparison is protocol-valid. *Disposition: deferred to future work* with a small resource ask ( $\sim 10$  GPU-minutes for a  $\leq 4$ -trial tuning-block sweep plus re-run; item 7)—not claim-critical, since a better-tuned baseline would likely strengthen the negative, and the under-tuned-spectral direction is already covered by the F12 wording.

4. **C5 era-identity probe inconclusive.** The probe of whether the kept subspace is era-specific found no significant gap in either direction at  $n=8$  probes (all Wilcoxon  $p \geq 0.078$ ). *Disposition: deferred*—a higher-power probe costs  $\sim 5$  GPU-minutes but is largely superseded by Theorem 4.1; carried as an open descriptive question (item 7).
5. **Stochastic eig non-convergence in the released implementation.** Under the rank-collapse regime this study documents, fp32 GPU eigendecomposition can fail to converge, crashing runs seed-dependently; the audit discovered this by re-execution, meaning the original 6-of-6 producer runs were lucky-complete. *Disposition: addressed by disclosure*—reported here as an implementation note; the audit’s mathematically equivalent CPU-fp64 fallback (which reproduced every conclusion) is preserved with the audit artifacts, and any follow-up must handle **eig** non-convergence.
6. **MLP corr-Sharpe CI is marginal.** The upper bound of the MLP corr-Sharpe difference CI straddles zero across bootstrap RNGs: producer  $[-0.416, -0.004]$ , auditor  $[-0.416, +0.003]$ . *Disposition: addressed by wording*—reported as marginal throughout; the robust GRU corr-Sharpe result ( $-0.242, [-0.414, -0.073]$ ) carries the stability-worsens statement.
7. **“Indistinguishable” means equivalence at  $\pm 0.004$ .** The C4 null is an equivalence claim at the resolution the design affords; spectral-vs-random differences smaller than  $\sim 0.002$ – $0.003$  cannot be excluded. *Disposition: addressed by wording* (stated with its CI wherever claimed); a tighter bound needs more seeds ( $\sim 6 \times 80$  s GPU per extra seed pair; item 7).
8. **The baseline-quartile tail split is descriptive only.** Bucketing eras on the baseline’s own realized corr induces regression to the mean mechanically; the monotone gradient in that table is expected artifactually. *Disposition: addressed by wording*—the unconditioned target-dispersion split is the evidentially meaningful breakdown (it shows uniform harm and no Feldman-tail pattern); a held-out-seed bucketing re-analysis of the existing CSVs is a free descriptive improvement.

Design-level limitations carried from the pre-registered triage:

- **Three seeds per arm (headline).** *Disposition: addressed*—the audit’s fresh-seed re-execution extends the MLP headline to a 6-seed pool ( $-0.00537$ , CI  $[-0.00853, -0.00222]$ ) with replication on an independently built shard (Section 4.7).
- **Single spectral operating point** (hard mode,  $mp\_factor$  2.0,  $B=1024$ ). A wider sweep inside this run would have broken the pre-registered matched-compute design. *Disposition: deferred to future work* (item 2, with re-registration).
- **The sequence arm is a reshaped-tabular construct** (F13): the GRU consumes a  $15 \times 47$  reshape of tabular features. The claim is architecture consistency of the optimizer effect, not sequence-modeling relevance. *Disposition: deferred*—a true second setting (OHL CV, both architectures) is item 3.
- **Single dataset, single loss.** Numerai v5 with MSE only; Theorem 4.1 is loss-specific, so other losses need new engagement diagnostics. *Disposition: deferred* (item 4). Context baselines (Muon, GBTs) were pre-designated cuts. *Disposition: attempted within the plan*

*but priced out*—kept as the designated cut under the 5-experiment budget; item 5. Feature-neutral correlation was not reported (a solid-tier extra,  $\sim 6$  GPU-minutes). *Disposition: deferred* (item 7).

- **MP-threshold null under cross-sectional correlation.** The MP bulk edge assumes i.i.d. samples; financial per-sample gradients are cross-sectionally correlated, so the threshold’s calibration on this data is empirical (via the C2 permutation control), not analytical. *Disposition: deferred* (item 6, CPU-only).
- **Accepted residual risk, stated in advance.** The pre-mortem priced a 1-in-4 to 1-in-3 chance of an unresolvable “could not determine”; that outcome did not occur (the F4 gate showed power 0.91, and a category was returned), but the neighboring risk it flagged—thresholds calibrated in a regime the budget does not deliver—partially materialized as the regime drift of limitation 1.

## 7 Future Work

Each item below is scoped precisely enough to seed a follow-up run: the experiment, the hypothesis it tests, how it builds on this round, and the resources required. The reference compute profile is the one this run used (a shared Slurm cluster’s free partition of NVIDIA L40 48 GB GPUs, jobs under 30 minutes); items exceeding it say so explicitly.

1. **Full-scale, non-subsampled Numerai verdict (removes the regime-drift scope limit).** Re-run the exp-004 protocol on the full training data and feature set at  $B \approx 4096$ , with F1/F5/F9 machinery re-registered on fresh eras. *Hypothesis:* the negative verdict persists when the baseline edge returns to the practitioner sanity band—i.e., the result is not an artifact of the low-edge subsampled regime (limitation 1). *Builds on:* the frozen protocol, thresholds-as-code, and the audit’s independent-shard tooling. *Resources:* est. 10–20 L40-hours, or A100-class GPUs for comfortable  $B=4096$  per-sample-gradient memory; **exceeds** the free-partition 30-minute cap—needs a paid or time-unconstrained allocation (order \$50–100 of cloud A100 time).
2. **Spectral operating-point sweep at a declared enlarged matched budget (closes the F12 asymmetry).** Give *both* arms a 12-trial budget (spectral:  $\text{mp\_factor} \times \text{mode} \times B$ ; AdamW re-swept at the same enlarged compute), pre-registered as a new protocol so the matched-compute design is redefined rather than broken. *Hypothesis:* no spectral configuration in a reasonable neighborhood of the prior-project priors recovers the baseline—upgrading “no evidence of benefit under the affordable tuning budget” toward “no benefit at any tested operating point”. *Builds on:* exp-003’s measured per-trial costs (63s/spectral run) and exp-001’s registered diagnostics. *Resources:*  $\sim 30$  GPU-minutes on one L40 (fits the free partition as two jobs); zero new data.
3. **True second setting: OHLCV dataset, both architectures (converts “architecture consistency” into “second setting”).** Build the pre-specified OHLCV protocol (public daily equities/crypto, purged temporal split,  $\geq 90\%$  coverage after cleaning) and run MLP and GRU arms with the identical verdict machinery. *Hypothesis:* the class-level harm of agreement-style filtering generalizes beyond Numerai’s obfuscated within-era construct to a natural sequential financial task. *Builds on:* exp-002’s validated vmap-GRU stack and exp-005’s within-era machinery. *Resources:*  $\sim 1\text{--}2$  L40-hours total across 2–4 jobs (fits the free partition); data is free to download;  $\sim 2\text{--}4$  hours of engineering for the split/cleaning protocol.

4. **Second loss family with target-sensitive engagement diagnostics (tests the theorem’s boundary).** Repeat the comparison with a corr-aligned per-era loss and/or a multi-output head, where Theorem 4.1 does not apply (the audit’s scoping controls confirm 2-output MSE breaks the invariance), using new engagement diagnostics (unnormalized-gradient spectra, residual-magnitude-weighted similarity, sign-pattern statistics of  $D$ ). *Hypothesis:* where the engagement spectrum is target-sensitive, the filter’s behavior (and possibly its effect) changes—directly testing whether target-blind selection is the operative harm mechanism. *Builds on:* the theorem and the C1 confound calibration. *Resources:*  $\sim 30$ – $60$  L40-minutes across 2–3 jobs (fits the free partition).
5. **Muon and GBT context baselines (the designated cuts).** Add a tuned Muon arm at matched budget [8] and a LightGBM-class GBT reference. *Hypothesis:* the tuned-AdamW baseline is not the weak link—the filter’s deficit persists against the current tabular bar. *Builds on:* the frozen t07 pipeline; pure arm additions. *Resources:*  $\sim 30$ – $60$  L40-minutes ( $\approx 2$  experiment slots); no new data.
6. **MP-null theory under cross-sectional correlation.** Simulation (and, where tractable, analytical) characterization of the Marchenko–Pastur bulk edge for per-sample gradients with financial factor structure, beyond the C2 permutation control—the shifted-edge question raised at design review. *Hypothesis:* the hard-mode threshold systematically misclassifies factor-driven directions as “signal” under realistic cross-sectional correlation, formalizing the C1(c) confound result. *Builds on:* the saved spectra (`spectra.npz`) and C1 test harness. *Resources:* CPU only—days of workstation CPU, no GPU, no cluster constraint.
7. **Small closure items** (bundle into one  $\sim 25$ -minute L40 job in any follow-up): (a) the  $\leq 4$ -trial GRU tuning sweep and exp-005 re-run ( $\sim 10$  GPU-minutes; limitation 3); (b) feature-neutral correlation reporting via saved per-row predictions ( $\sim 6$  GPU-minutes); (c) a higher-power C5 era-identity probe ( $\sim 5$  GPU-minutes); (d) extra seed pairs to tighten the C4 equivalence bound below  $\pm 0.002$  ( $\sim 80$ s per seed pair); and (e) the free held-out-seed re-bucketing of the tail-era split (CPU-only re-analysis of existing CSVs).

## 8 Conclusion

We asked whether a Marchenko–Pastur spectral filter on per-sample gradients—a mechanism with strong label-noise credentials in classification—reduces the effect of noise when training on large, low-SNR financial time-series data. Under a fully pre-registered protocol with era-purged splits, frozen thresholds, block-bootstrap inference, and compute-matched tuning, the answer is negative: no evidence of benefit under the affordable tuning budget, with the filter significantly degrading out-of-sample performance at the pre-registered operating point on both an MLP and a GRU, a result that survived independent fresh-seed re-execution on an independently constructed evaluation split.

The run’s equally valuable output is the explanation. For scalar-output MSE, the filter’s engagement eigenspectrum is provably target-independent: “consensus” measures feature/Jacobian factor structure, not signal, so the coherence-amplifier mechanism that worked under label noise has nothing to select for in this regime. Empirically, MP-eigenselection was statistically indistinguishable from a norm- and  $k$ -matched random subspace projection, and all agreement-style variants hurt similarly—the harm belongs to the intervention class, not the spectral machinery. Together these results delimit where coherence-amplifying optimizers can be expected to apply: the mechanism requires a loss whose per-sample gradient geometry is target-sensitive, and a regime where coherent target signal is separable from factor structure—conditions the low-SNR financial regression setting, by proof and by measurement, does not provide.

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